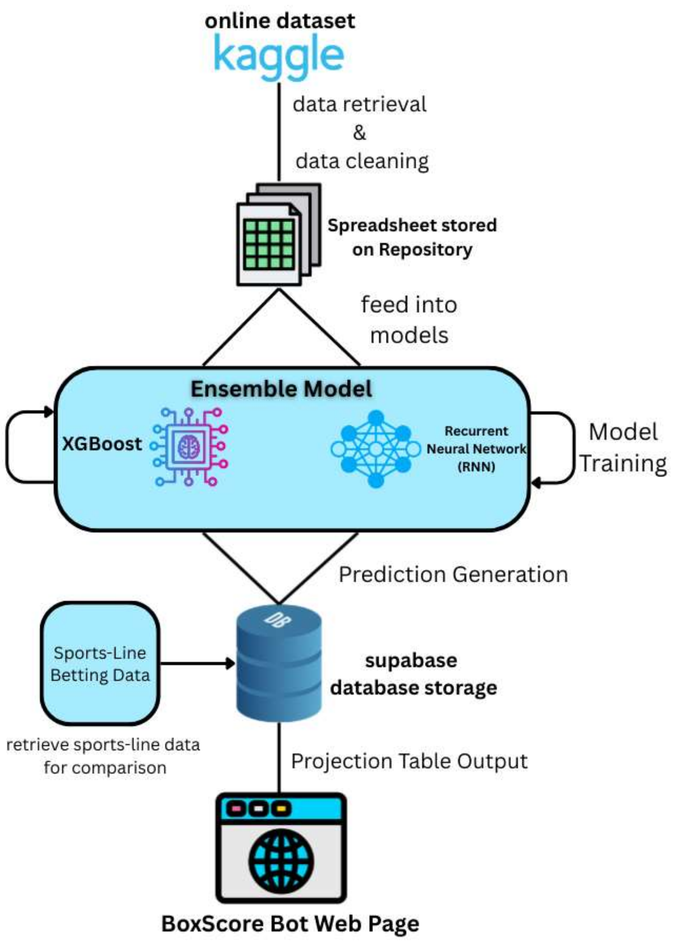
University of CINCINNATI
CEAS

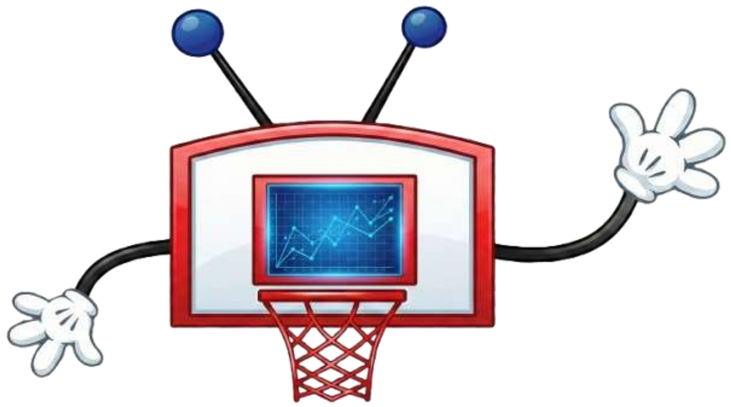
Box Score Bot

How It Works



Abstract

Box Score Bot is a Machine Learning powered sports line predictor. This project utilizes public statistics on player and team performances and uses these statistics to generate projections on how these players will perform in the future based on underlying trends in the data and upcoming matchup statistics. These predictions are integrated and hosted on a website which will allow ease-of-use and accessibility. We also include a projection ranking page which pulls existing projections namely from sports-betting platforms to compare and advise on what we believe is a "good" bet.



Goals

This project was started as a way to generate valuable and accurate projections for NBA player statistics such as points, rebounds, and assists. This can be extremely useful due to the nature of sportsbooks which simply set lines at where people will bet both sides, not where the actual projection for players is. Our models allow us to find what we believe to be bad lines and make recommendations based off that. This project could also provide value to team managers who need to decide which player will provide more team value in a given game.

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Challenges

Data Source

- Finding a reliable data source was initially difficult, with some requiring purchasing tokens to access stats, and others not up to date with the current season's games and active players
- We eventually found a good dataset from Kaggle, containing all the player statistics we need to properly train our models, and it would continuously update with new game stats throughout the season

API

- The API for tracking active NBA players this season was also a challenge to work with, since there was an incident where someone outside changed the logic of one of the API getter functions, temporarily breaking our project
- We eventually were able to work around this and revert the logic back, but it was a necessary warning sign that we should account for sudden API changes that could stop our app from functioning.

Model Accuracy

- The current challenge we're working on is improving the model accuracy, which is currently reaching only 50% accuracy, which isn't very helpful on indicating how a player will improve over the season.
- We are investigating new training techniques and data cleaning tweaks to push our models to their fullest potential.

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